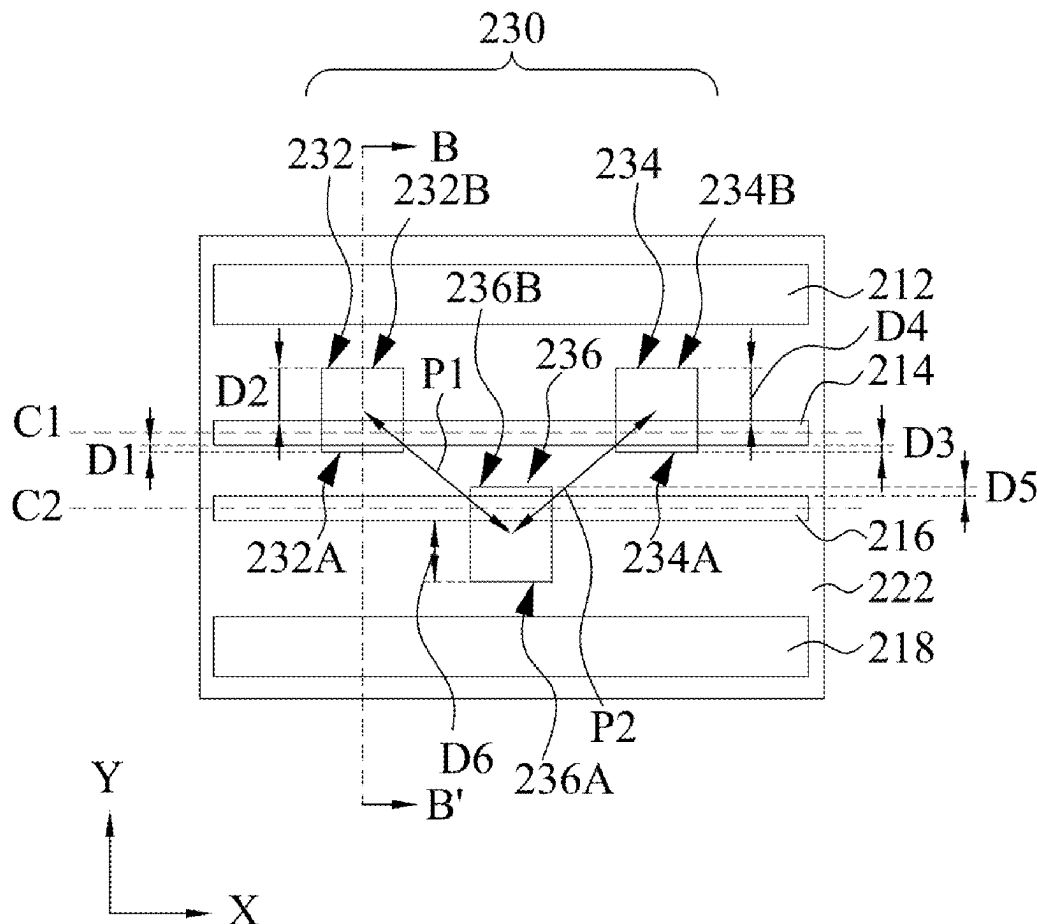




US 20250285877A1

(19) **United States**(12) **Patent Application Publication**
LIN et al.(10) **Pub. No.: US 2025/0285877 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **FEATURE PATTERNING USING PITCH
RELAXATION AND DIRECTIONAL
END-PUSHING WITH ION BOMBARDMENT**(60) Provisional application No. 63/085,545, filed on Sep.
30, 2020.**Publication Classification**(71) Applicant: **Taiwan Semiconductor
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(TW)(51) **Int. Cl.**
H01L 21/3115 (2006.01)
H01J 37/317 (2006.01)(72) Inventors: **Tzung-Hua LIN**, Hsinchu (TW); **Yi-Ko
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(TW)(52) **U.S. Cl.**
CPC **H01L 21/31155** (2013.01); **H01J 37/3171**
(2013.01); **H01J 2237/3171** (2013.01)(73) Assignee: **Taiwan Semiconductor
Manufacturing Co., Ltd.**, Hsinchu
(TW)(57) **ABSTRACT**

A method for forming a semiconductor structure includes forming a pattern having first and second line features extending in a first direction on a substrate. After depositing a photoresist layer on the substrate to cover the pattern, the photoresist layer is patterned to form a cut pattern including first and second cut features exposing portions of the respective first and second line features. In a top view, at least one of the first and second cut features is asymmetrically arranged with respect to a central axis of a corresponding first or second line feature. At least one angled ion implantation is performed to enlarge the first and second cut features in at least one direction perpendicular to the first direction. The portions of the first and second line features exposed by the respective first and second cut features are then removed.

(21) Appl. No.: **19/214,861**(22) Filed: **May 21, 2025****Related U.S. Application Data**(63) Continuation of application No. 18/325,898, filed on
May 30, 2023, now Pat. No. 12,315,737, which is a
continuation of application No. 17/191,541, filed on
Mar. 3, 2021, now Pat. No. 11,688,610.

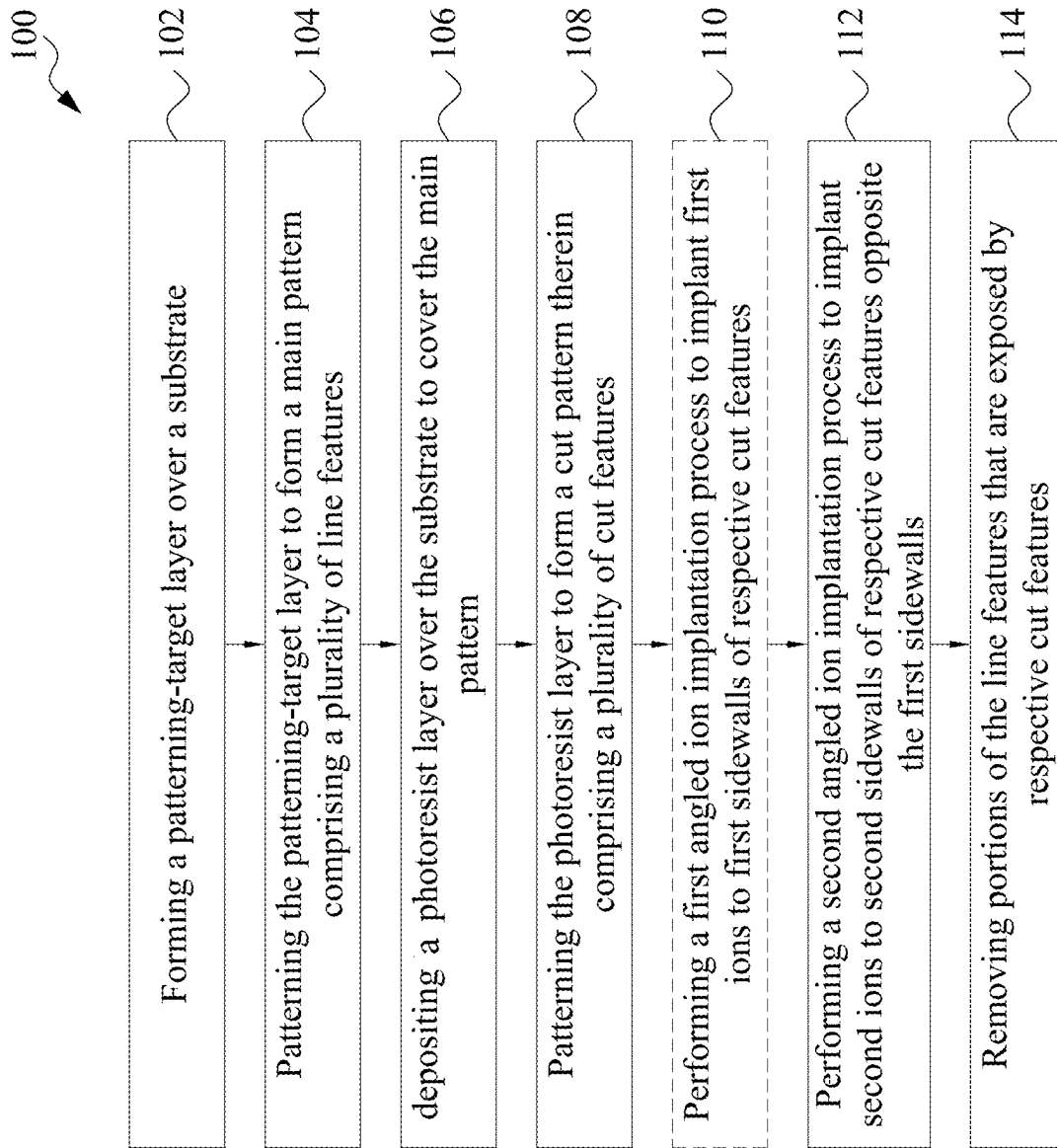


FIG. 1

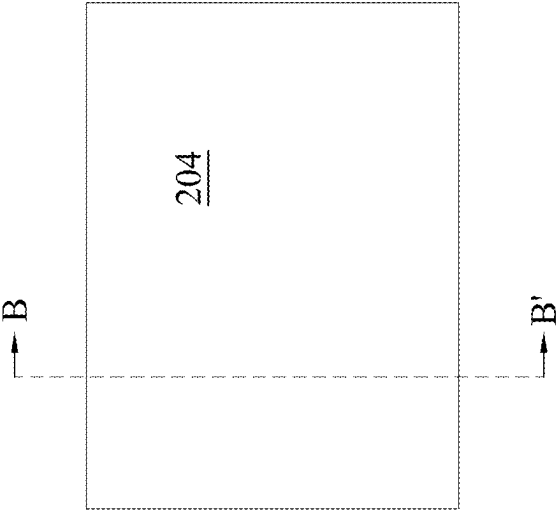


FIG. 2A

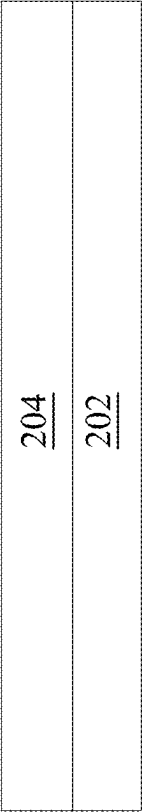


FIG. 2B

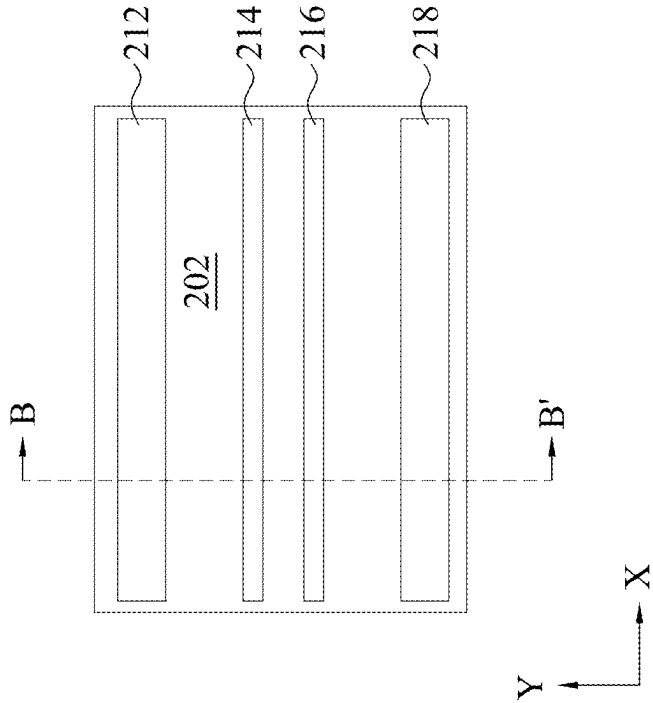


FIG. 3A

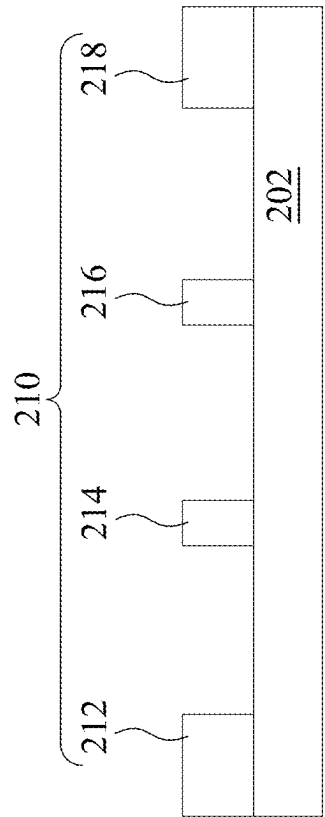


FIG. 3B

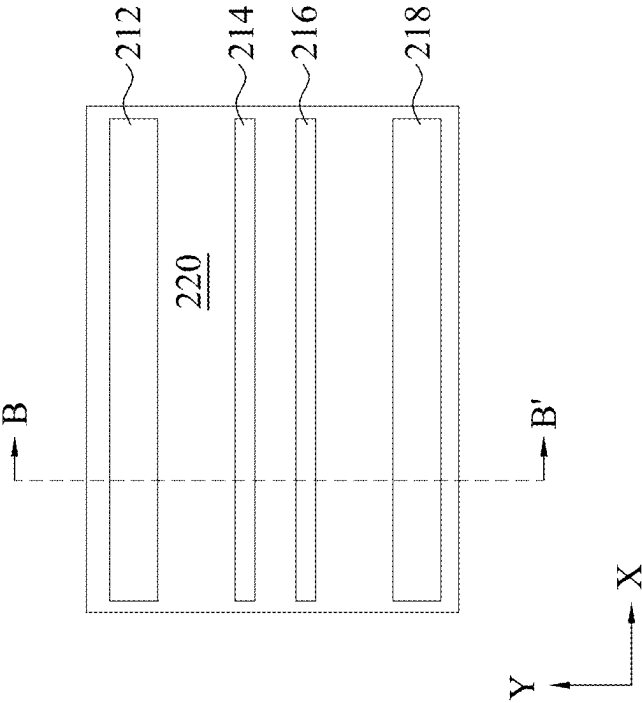


FIG. 4A

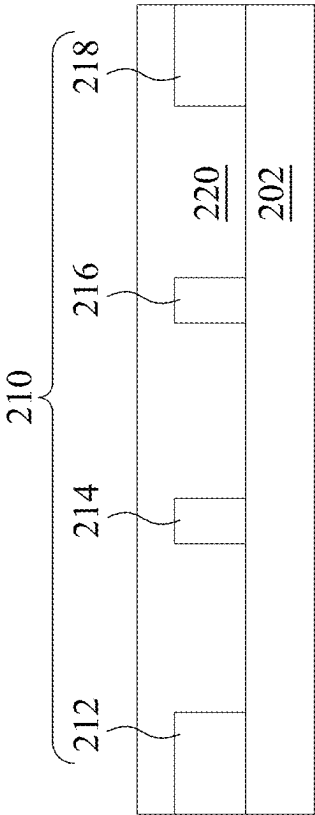


FIG. 4B

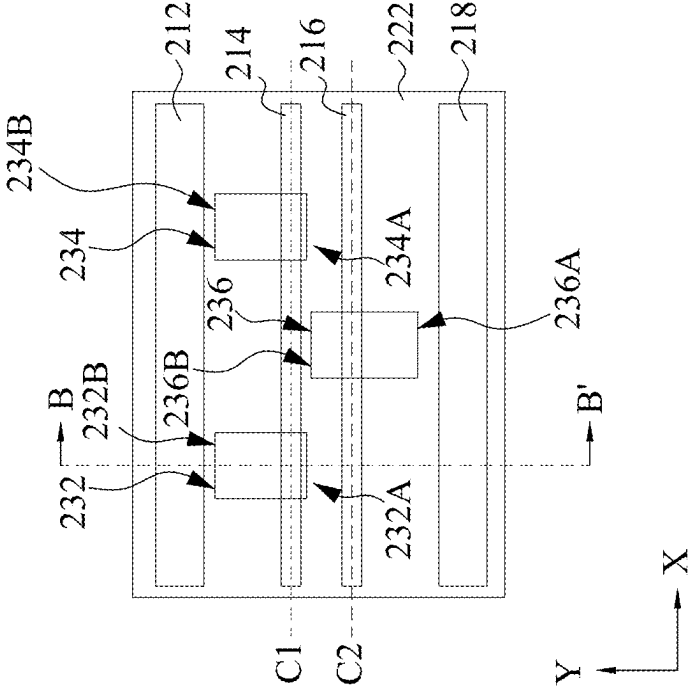


FIG. 6A

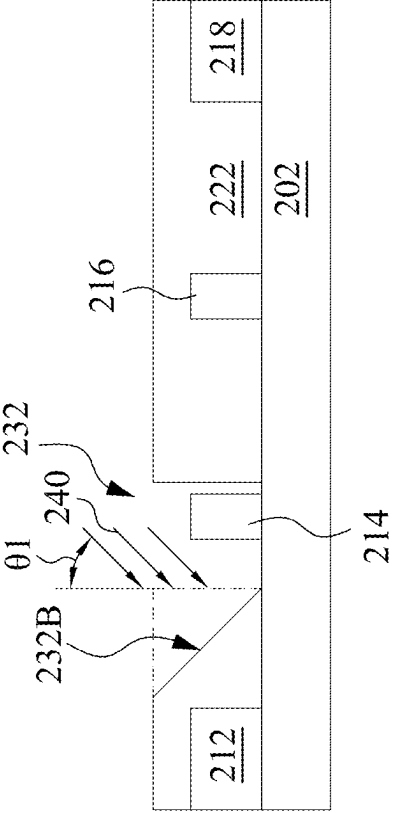


FIG. 6B

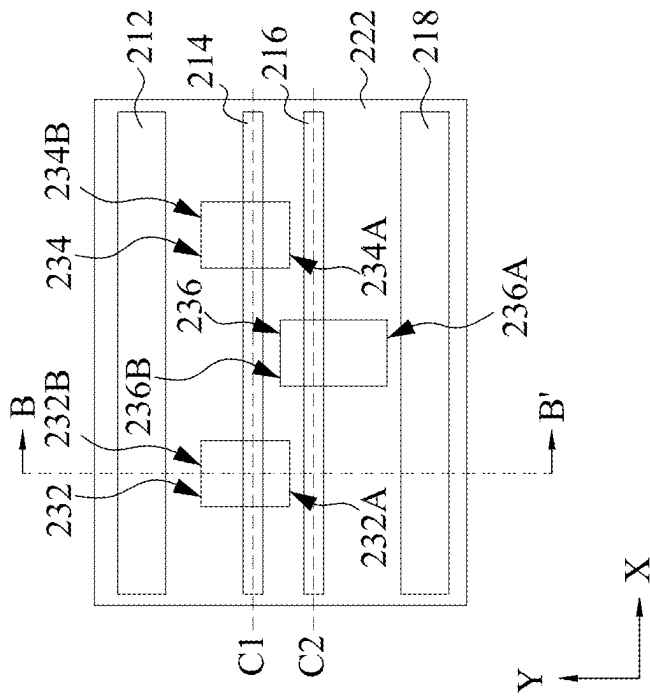


FIG. 7A

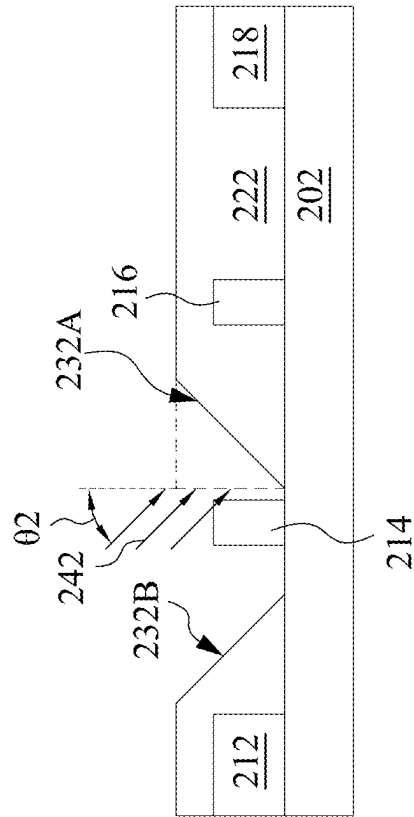


FIG. 7B

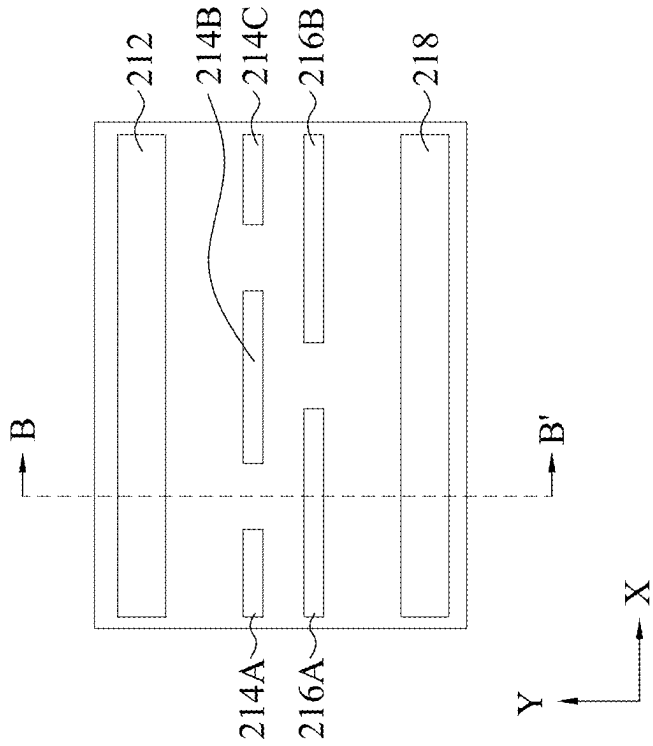


FIG. 8A

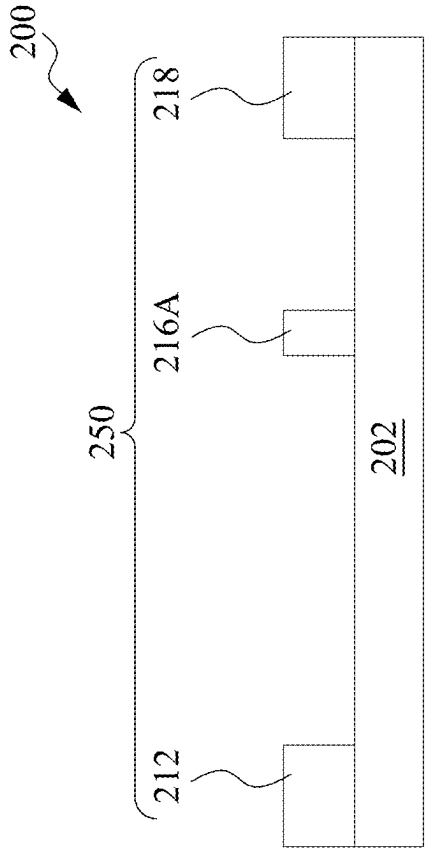


FIG. 8B

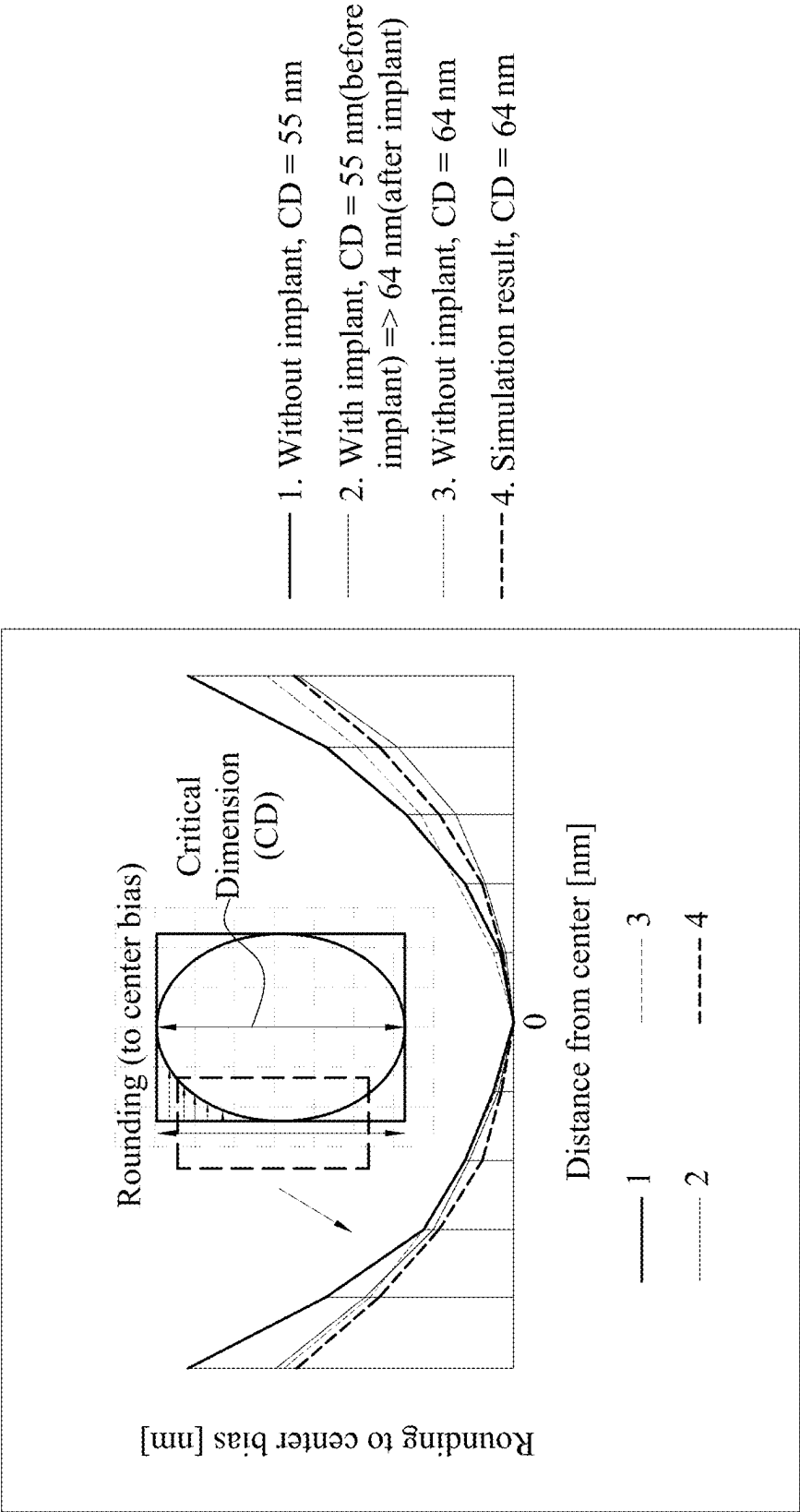


FIG. 9

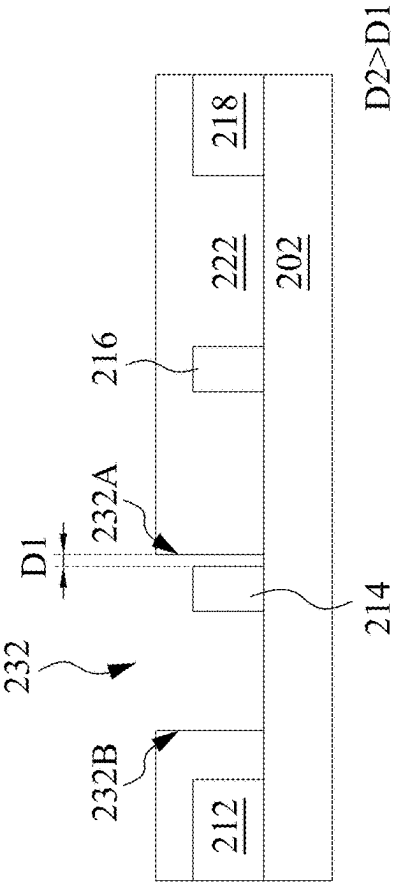


FIG. 10B

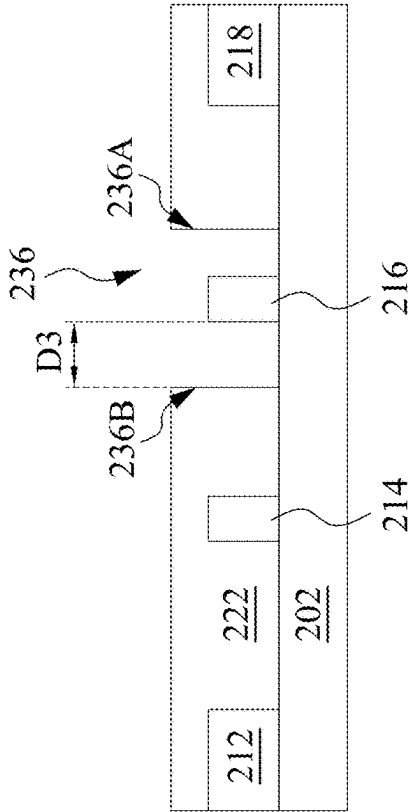


FIG. 10C

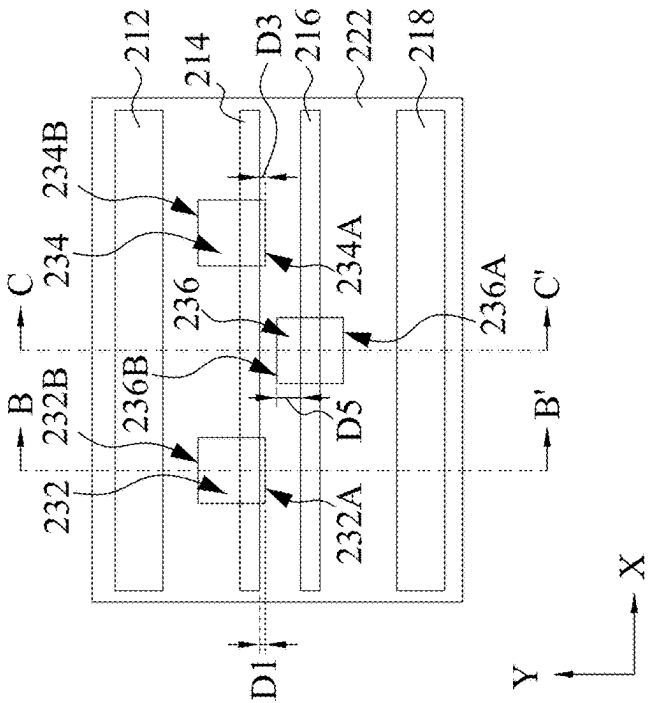


FIG. 10A

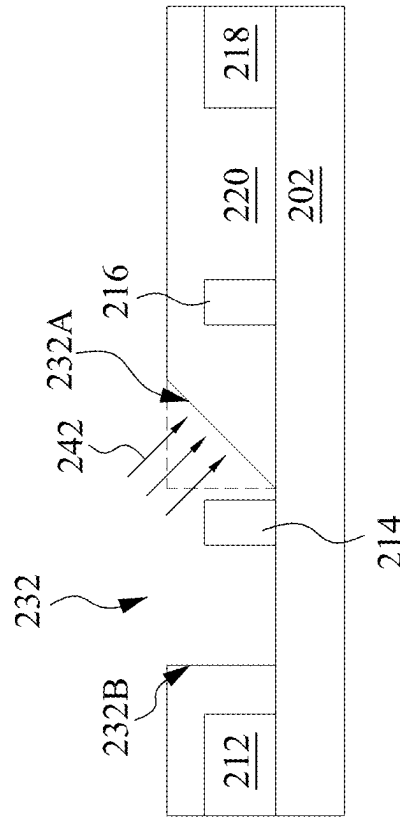


FIG. 11B

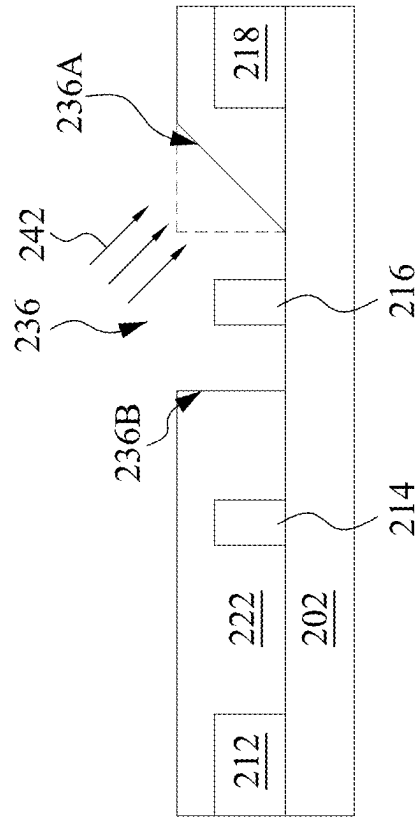


FIG. 11C

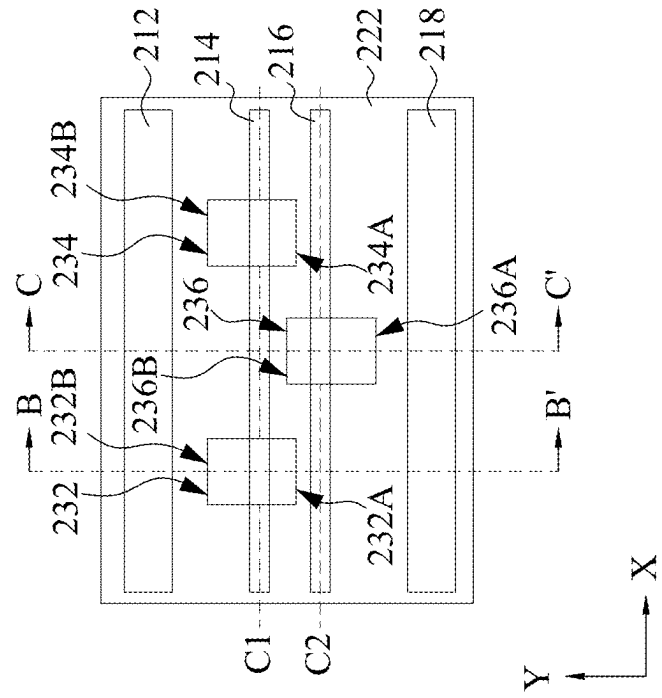
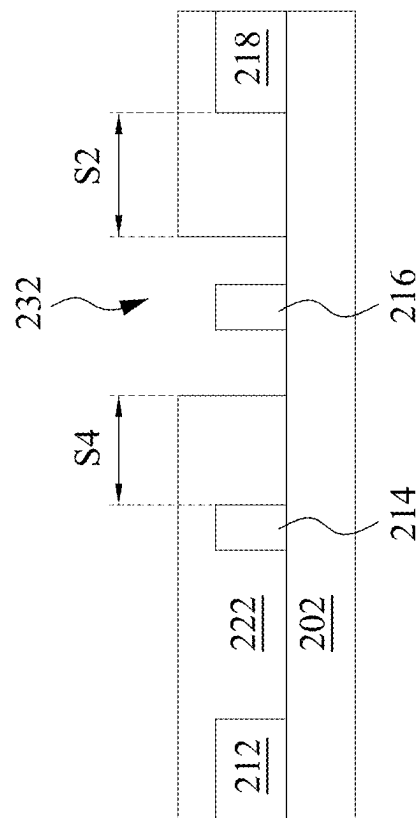
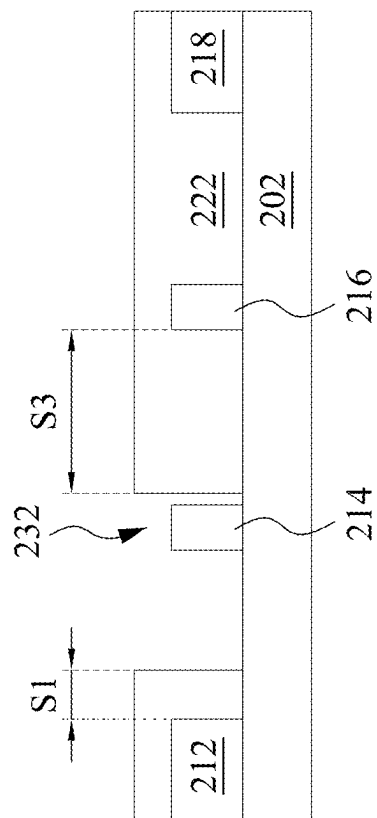
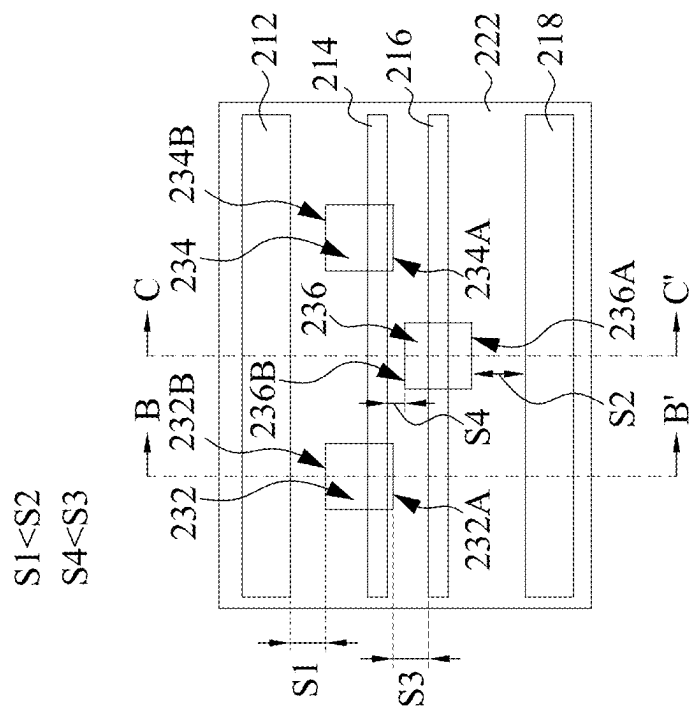


FIG. 11A



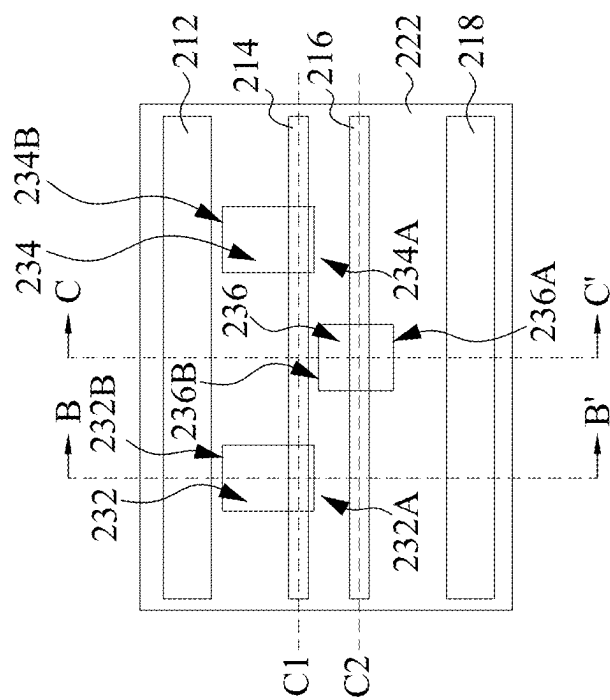
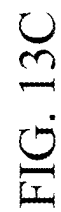
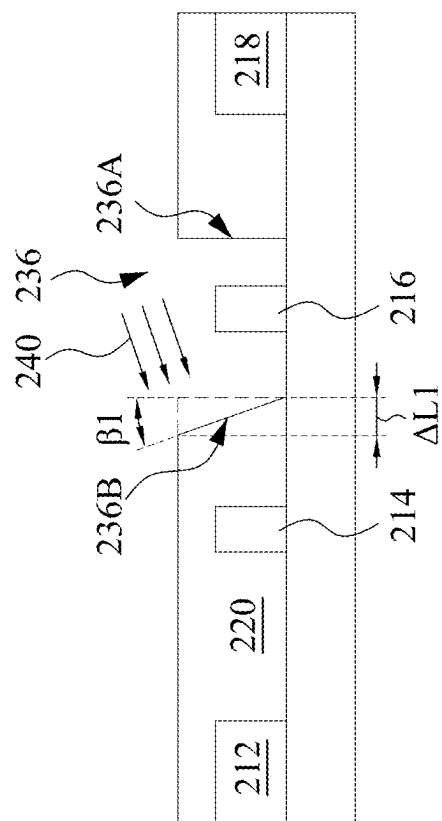
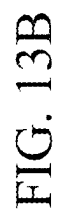
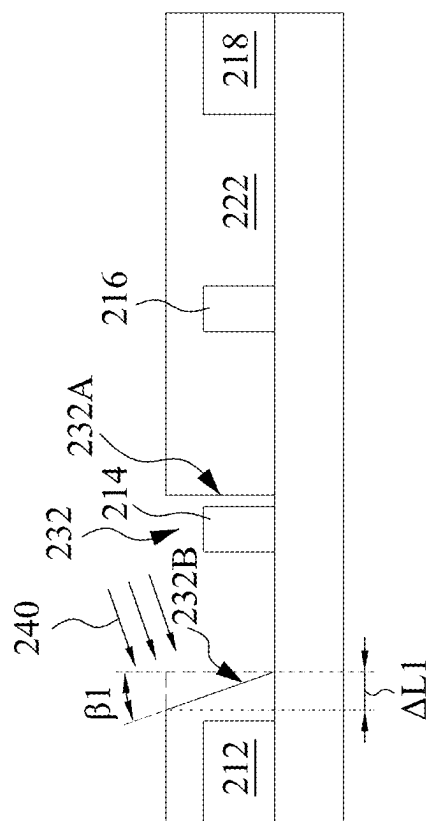
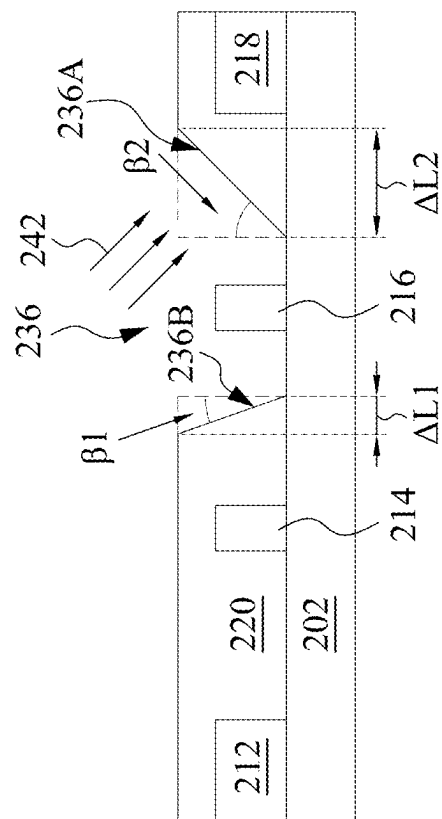
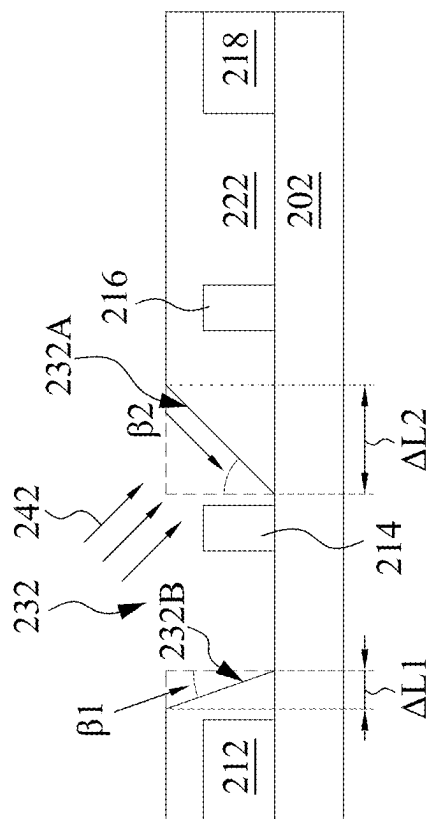
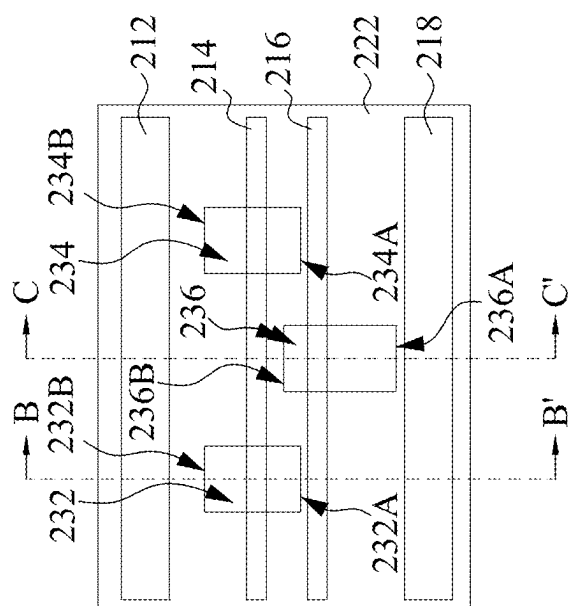


FIG. 13A



FEATURE PATTERNING USING PITCH RELAXATION AND DIRECTIONAL END-PUSHING WITH ION BOMBARDMENT

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 18/325,898, filed May 30, 2023, which is a continuation of U.S. patent application Ser. No. 17/191,541, filed Mar. 3, 2021, which claims the benefit of priority to U.S. Provisional Application No. 63/085,545, filed Sep. 30, 2020, which applications are incorporated by reference herein in their entireties.

BACKGROUND

[0002] Integrated circuit (IC) technologies are constantly being improved. Such improvements frequently involve scaling down device geometries to achieve lower fabrication costs, higher device integration density, higher speeds, and better performance. Lithography is frequently used for forming components of an integrated circuit device, where generally, an exposure tool passes light through a photomask and focuses the light onto a photoresist layer over a substrate, resulting in the photoresist layer having an image of integrated circuit components therein. Forming device patterns with smaller dimensions is limited by a resolution of the exposure tool.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a flow chart of an example method for fabricating a semiconductor structure, in accordance with some embodiments.

[0005] FIGS. 2A-8B are various views of the semiconductor structure at various fabrication stages constructed according to the method of FIG. 1, in accordance with some embodiments.

[0006] FIG. 9 illustrates effects of ion implantation on the pattern profile for exemplary cut features.

[0007] FIGS. 10A-11C are various views of the semiconductor structure at various fabrication stages constructed according to the method of FIG. 1, in accordance with first alternative embodiments.

[0008] FIGS. 12A-14C are various views of the semiconductor structure at various fabrication stages constructed according to the method of FIG. 1, in accordance with second alternative embodiments.

DETAILED DESCRIPTION

[0009] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct

contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0010] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0011] Integrated circuits may be formed using various photolithographic techniques. Such techniques involve exposing a photoresist layer to a light source through a photomask, thereby forming a pattern in the photoresist layer. The regions where the photoresist layer is removed expose the underlying substrate to an etching process used to transfer the pattern into the underlying substrate. As the patterns formed into the photoresist layer become increasingly dense, it becomes difficult to use a single photomask to form a pattern in the photoresist layer because features are smaller than the resolution of a light source to which the photoresist layer is exposed. Thus, multiple masks may be used to form the features within a pattern.

[0012] In some cases, a target pattern is formed through use of both a main pattern and a cut pattern. The cut pattern is used to remove features formed by the main pattern from the unwanted areas in order to achieve the desired target pattern. As the density of features in the main pattern increases, some cut features in the cut pattern may be too close to one another, thus a single photomask would not be able to form such a cut pattern when an immersion lithography system is used. In some approaches, multiple photomasks are used to form the cut pattern. Using multiple photomasks, however, increases process errors and fabrication cost. The extreme ultraviolet (EUV) lithography is known to extend the range of lithography processes to smaller feature sizes and pitches. However, EUV lithography requires very sophisticated reflecting optics or mirrors, which significantly increases fabrication cost.

[0013] In embodiments of the present disclosure, the lithography printability for cut features with small pitches is enlarged by a pitch relaxation process in which some cut features are shifted away to increase the distance therebetween, thereby allowing the cut features to be printed using a single photomask. The sizes of some cut features are then increased by ion implantation to recover the cut margin, thereby ensuring a sufficient cut window. The embodiments of the present disclosure advantageously reduce the fabrication cost and processing errors.

[0014] FIG. 1 is a flowchart of a method 100 for fabricating a semiconductor structure 200, in accordance with some embodiments. FIGS. 2A-8B are various views of the semiconductor structure 200 in various stages of the method 100, in accordance with some embodiments. The method 100 is discussed in detail below, with reference to the

semiconductor structure **200**, in FIGS. 2A-8B. In some embodiments, additional operations are performed before, during, and/or after the method **100**, or some of the operations described are replaced and/or eliminated. In some embodiments, additional features are added to the semiconductor structure **200**. In some embodiments, some of the features described below are replaced or eliminated. One of ordinary skill in the art would understand that although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

[0015] Referring to FIGS. 1 and 2A-2B, the method **100** includes operation **102**, in which a patterning-target layer **204** is formed over a substrate **202**. FIG. 2A is a top view of a semiconductor structure after forming the patterning-target layer **204** over the substrate **202**, in accordance with some embodiments. FIG. 2B is a cross-sectional view of the semiconductor structure of FIG. 2A taken along line B-B'.

[0016] Referring to FIGS. 2A and 2B, in some embodiments, the substrate **202** is a bulk semiconductor substrate including a semiconductor material or a stack of semiconductor materials such as, for example, silicon (Si), germanium (Ge), silicon germanium (SiGe), carbon doped silicon (Si:C), silicon germanium carbon (SiGeC); or an III-V compound semiconductor such as, for example, gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), indium antimonide (InSb), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), or gallium indium arsenide phosphide (GaInAsP). In some embodiments, the bulk semiconductor substrate includes a single crystalline semiconductor material such as, for example, single crystalline silicon. In some embodiments, the substrate **202** is a semiconductor-on-insulator (SOI) substrate including a top semiconductor layer formed on an insulator layer (not shown). The top semiconductor layer includes the above-mentioned semiconductor material such as, for example, Si, Ge, SiGe, Si:C, SiGeC; or an III-V compound semiconductor including GaAs, GaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, or GaInAsP. The insulator layer is, for example, a silicon oxide layer, or the like. The insulator layer is provided over a base substrate, typically a silicon or glass substrate. The substrate **202** may further include additional features and/or material layers, such as various isolation features formed in the substrate. In some embodiments, the substrate **202** may include various doped regions, such as p-type doped regions and/or n-type doped regions configured and coupled to form various devices and functional features. The term “p-type” refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. Exemplary p-type dopants, i.e., p-type impurities, include, but are not limited to, boron, aluminum, gallium, and indium. The term “n-type” refers to the addition of impurities that contribute free electrons to an intrinsic semiconductor. Exemplary n-type dopants, i.e., n-type impurities, include, but are not limited to, antimony, arsenic, and phosphorus. If doped, the substrate **202**, in some embodiments, has a dopant concentration in a range from 1.0×10^{14} atoms/cm³ to 1.0×10^{17} atoms/cm³, although the dopant concentrations may be greater or smaller. All doping features may be achieved using a suitable process, such as ion implantation in various steps and techniques. In

some embodiments, the substrate **202** may include other features, such as shallow trench isolation (STI).

[0017] The patterning-target layer **204** is formed over the substrate **202**. In some embodiments, the patterning-target layer **204** is the layer where the final patterns are formed over the substrate **202**. In some embodiments, the patterning-target layer **204** has a thickness in a range from about 5 nm to about 50 nm. If the patterning-target layer **204** is too thin, it cannot be deposited as a uniform layer. If the patterning-target layer **204** is too thick, it requires a long etching time, which increases fabrication cost. In some embodiments, the patterning-target layer **204** is formed using a deposition process such as, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), atomic layer CVD (ALCVD), or spin coating. In some embodiments, the patterning-target layer **204** is formed by converting a surface portion of the substrate **202** using a thermal oxidation or a nitridation process. In some embodiments, the patterning-target layer **204** includes one or more dielectric materials, such as silicon oxide (SiO₂), and/or silicon nitride (Si₃N₄). In some embodiments, the patterning-target layer **204** includes one or more metallic materials. In some embodiments, the patterning-target layer **204** is an upper portion of the substrate **202** and includes one or more semiconductor materials.

[0018] Referring to FIGS. 1 and 3A-3B, the method **100** proceeds to operation **104**, in which the patterning-target layer **204** is patterned to form a main pattern **210**. FIG. 3A is a top view of the semiconductor structure of FIGS. 2A and 2B after patterning the patterning-target layer **204** to form the main pattern **210** over the substrate **202**, in accordance with some embodiments. FIG. 3B is a cross-sectional view of the semiconductor structure of FIG. 3A taken along line B-B'.

[0019] Referring to FIGS. 3A and 3B, the main pattern **210** includes a plurality of line features **212**, **214**, **216**, **218** extending in a first direction, for example, an X direction. The line features **212**, **214**, **216**, **218** are spaced apart from each other in a second direction, for example, a Y direction that is different from the first direction, the X direction. In some embodiments, the first direction (i.e., X direction) is a horizontal direction, and the second direction (i.e., Y direction) is a vertical direction. Although four line features **212**, **214**, **216**, **218** are shown and described in FIG. 3A, any numbers of line features are contemplated. Further, although the line features **212**, **214**, **216**, **218** are illustrated as rectangular-shaped lines, such is not required for some embodiments and any shapes of line features are contemplated. In some embodiments, line features **212**, **214**, **216**, **218** are dummy features and will be removed at a later fabrication stage. In some embodiments, line features **212**, **214**, **216**, **218** are functional features, such as line features or metal lines.

[0020] The line features **212**, **214**, **216**, **218** may be formed to have a same or different width(s). In some embodiments, the line features **212**, **214**, **216**, **218** are formed corresponding to, for example, active regions in a static random access memory (SRAM) cell. Accordingly, the outer line features, i.e., line features **212** and **218**, are formed to be wider than the inner line features, i.e., line features **214** and **216**, and the spacing between two inner line features, i.e., line features **214**, **216**, is smaller than the spacing between an inner line feature, i.e., line feature **214** or **216**, and a corresponding

adjacent outer line feature, i.e., line feature **212** or **218**. In some embodiments, each of the outer line features **212**, **218** has a width of 15.5 nm, and each of the inner line features **214**, **216** has a width of 8 nm. In some embodiments, the spacing between the inner line features **214**, **216** is about 32 nm, and the spacing between the inner line feature **216** and the outer line feature **218** is about 39 nm.

[0021] As integrated circuit features continue to shrink in size, the pitches and critical dimensions of line features **212**, **214**, **216**, **218** also shrink. In some embodiments, the pitch of adjacent inner line features, i.e., line features **214** and **216**, is between 20 nm and 40 nm. As used here, the term “pitch” is the distance between identical points in two neighboring features. In some instances, pitch can be viewed as the distance from a center of one feature to a center of another neighboring feature.

[0022] The patterning-target layer **204** may be patterned by any suitable method. In some embodiments, the patterning-target layer **204** is patterned using a single lithography process. In some embodiments, a first photoresist layer (not shown) is applied over the patterning-target layer **204**. In some embodiments, the first photoresist layer includes chemicals that are sensitive to light, such as UV light. In some embodiments, the first photoresist layer includes one or more organic polymer materials. In some embodiments, the deposition of the first photoresist layer includes a spin coating process and may be followed by a baking process. In some embodiments, the first photoresist layer is formed to have a thickness in a range from about 30 nm to about 100 nm, although lesser or greater thicknesses can also be employed.

[0023] The first photoresist layer is subsequently patterned using a lithography process. In some embodiments, patterning the first photoresist layer includes exposing the first photoresist layer to a radiation, post-exposure baking, developing the first photoresist layer in a resist developer, and hard baking thereby removing exposed portions in the case of positive resist (or unexposed portions in the case of negative resist) of the first photoresist layer, leaving unexposed portions thereof on the patterning-target layer **204** as the resist pattern. The resist pattern corresponds to the main pattern **210** to be formed in the patterning-target layer **204**. In some embodiments, the lithography process may alternatively employ other suitable technology, such as electron-beam direct writing.

[0024] The patterned first photoresist layer is then used as an etching mask to transfer the pattern in the first photoresist layer to the patterning-target layer **204**. In some embodiments, the regions of the patterning-target layer **204** that are not covered by the patterned first photoresist layer are removed using one or more anisotropic etching processes, leaving the regions corresponding to the main pattern **210** remain in the patterning-target layer **204**. In some embodiments, the one or more anisotropic etching processes include a dry etch process, such as a plasma etching process or a reactive ion etching (RIE) process, a wet etching process, or a combination thereof. In some embodiments, the anisotropic etching process includes using etching gases including at least one of carbon tetrafluoride (CF_4), difluoromethane (CHF_2), trifluoromethane (CHF_3), other suitable etching gases, or combinations thereof.

[0025] After forming the main pattern **210** in the patterning-target layer **204**, the patterned first photoresist layer is removed. In some embodiments, the patterned first photo-

resist layer is removed by a wet stripping process, a plasma ashing process, other suitable methods, and/or combinations thereof. In some embodiments, the plasma ashing process includes using gases including at least one of oxygen (O_2), carbon dioxide (CO_2), nitrogen (N_2), hydrogen (H_2), or combinations thereof.

[0026] In some embodiments, the patterning-target layer **204** is patterned using two or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over the patterning-target layer **204** and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the patterning-target layer **204** to form the main pattern **210**.

[0027] Referring to FIGS. 1 and 4A-4B, the method **100** proceeds to operation **106**, in which a second photoresist layer **220** is deposited over the substrate **202** to cover the main pattern **210**. FIG. 4A is a top view of the semiconductor structure of FIGS. 3A and 3B after depositing the second photoresist layer **220** over the substrate **202** to cover the main pattern **210**, in accordance with some embodiments. FIG. 4B is a cross-sectional view of the semiconductor structure of FIG. 4A taken along line B-B'.

[0028] Referring to FIGS. 4A and 4B, the second photoresist layer **220** is formed over the main pattern **210** and the substrate **202**. In some embodiments, the second photoresist layer **220** includes chemicals that are sensitive to light, such as UV light. In some embodiments, the second photoresist layer **220** includes one or more organic polymer materials. In some embodiments, the second photoresist layer **220** includes materials that are substantially similar to the materials of the first photoresist layer. In some embodiments, the deposition of the second photoresist layer **220** includes a spin-on coating process and may be followed by a baking process. In some embodiments, the second photoresist layer **220** is formed to completely fill the spaces between the line features **212**, **214**, **216**, **218** of the main pattern **210**. Accordingly, a top surface of the second photoresist layer **220** is located above top surfaces of the line features **212**, **214**, **216**, **218**. In some embodiments, the second photoresist layer **220** has a thickness in a range from about 10 nm to about 100 nm, although lesser or greater thicknesses can also be employed.

[0029] Referring to FIGS. 1 and 5A-5B, the method **100** proceeds to operation **108**, in which the second photoresist layer **220** is patterned to form a patterned second photoresist layer **222** comprising a cut pattern **230**. FIG. 5A is a top view of the semiconductor structure of FIGS. 4A and 4B after patterning the second photoresist layer **220** to form the patterned second photoresist layer **222** comprising the cut pattern **230**, in accordance with some embodiments. FIG. 5B is a cross-sectional view of the semiconductor structure of FIG. 5A taken along line B-B'.

[0030] Referring to FIGS. 5A and 5B, the cut pattern **230** includes a plurality of cut features, e.g., cut features **232**, **234**, **236**, overlying the main pattern **210**. The cut features **232**, **234**, **236** are intended to remove unnecessary portions

of the line features **212**, **214**, **216**, **218** so that the circuit associated with the resulting target pattern will function properly. In some embodiments, the cut features **232**, **234**, **236** are formed to expose portions of inner line features, i.e., line features **214**, **216**, to be removed in the following operations according to design specification. In some embodiments, the cut features **232**, **234**, **236** are trenches extending through the second photoresist layer **220**. In some embodiments, the cut pattern **230** includes a first cut feature **232** that exposes a first portion of the line feature **214**, a second cut feature **234** that is adjacent to the first cut feature **232** and exposes a second portion of the line feature **214**, and a third cut feature **236** exposing a portion of the line feature **216**. Although three cut features **232**, **234**, **236** are shown and described in FIG. 5A, any numbers of cut features are contemplated in the present disclosure. As shown in FIG. 5A, the first cut feature **232** includes opposite sidewalls **232A**, **232B** with sidewall **232A** extending into the space between the closely spaced line features **214** and **216** and sidewall **232B** extending into the space between the widely spaced line features **212** and **214**. The second cut feature **234** includes opposite sidewalls **234A**, **234B** with sidewall **234A** extending into the space between the closely spaced line features **214** and **216** and sidewall **234B** extending into the space between the widely spaced line features **212** and **214**. The third cut feature **236** includes opposite sidewalls **236A**, **236B** with sidewall **236A** extending into the space between the closely spaced line features **214** and **216** and sidewall **236B** extending into the space between the widely spaced line features **216** and **218**.

[0031] In some embodiments, a single, direct lithography process is performed to pattern the second photoresist layer **220** to form the cut features **232**, **234**, **236**. In some embodiments, the lithography process includes exposing the second photoresist layer **220** to a light source using a photomask, performing post-exposure bake processes, and developing the second photoresist layer **220** to form the cut features **232**, **234**, **236** in the second photoresist layer **220**. The cut features **232**, **234**, **236** may be formed in any shape such as a square, rectangular, round, or oval shape. In some embodiments and as in FIG. 5B, the cut features **232**, **234**, **236** are formed with substantially vertical sidewalls. In some other embodiments, the cut features **232**, **234**, **236** are formed with inclined sidewalls (not shown). In some embodiments, the cut features **232**, **234**, **236** are formed to have a same size. In some embodiments, the cut features **232**, **234**, **236** are formed to have different sizes.

[0032] As with all features in integrated circuits, the cut features **232**, **234**, **236** have to conform to design rules and other constraints. In order to be formed with a single exposure of a single photomask at a given technology node, thus eliminating the need of using multiple photomasks or the expensive EUV lithography technique, the pitch between each adjacent pair of cut features **232**, **234**, **236** has to be greater than the minimum pitch that can be obtained with a single lithography process at the given technology node. For example, the first pitch P1 between the first cut feature **232** and the third cut feature **236** and the second pitch P2 between the second cut feature **234** and the third cut feature **236** all need to be greater than the minimum pitch of the single lithography process. To provide an adequate lithography window, the cut features **232**, **234**, **236** are formed in a relaxed-pitch arrangement with a comparatively large pitch. This can be done by moving at least one cut feature

232, **234** and/or **236** in the cut pattern **230** away from along the Y direction so as to increase the distances between the adjacent cut features **232**, **234**, **236**. The pitch P1 and/or P2 between adjacent cut features **232**, **234**, **236** can thus be increased, allowing forming the cut features **232**, **234**, **236** by a single lithography process using a single photomask. In some embodiments and as shown in FIG. 5A, the cut features **232**, **234** are moved toward the outer line feature **212** along the positive Y direction, while the cut feature **236** is moved toward the outer line feature **218** along the negative Y direction. As a result, the pitch P1 between the cut features **232** and **236** and the pitch P2 between the cut features **234** and **236** are relaxed.

[0033] As a result of the pitch relaxation, the cut features **232**, **234**, **236** are asymmetrically arranged with respect to respective central axes of the line features **214**, **216**. For example, the first cut feature **232** is asymmetrically arranged with respect to a central axis C1 of the line feature **214** extending along the X direction such that the sidewall **232A** of the first cut feature **232** is positioned closer to the line feature **214** than the opposite sidewall **232B**. Accordingly, the spacing D1 between the sidewall **232A** of the first cut feature **232** and the line feature **214** is smaller than the spacing D2 between the opposite sidewall **232B** of the first cut feature **232** and the line feature **214**. Similarly, the second cut feature **234** is asymmetrically arranged with respect to the central axis C1 of the line feature **214** such that the sidewall **234A** of the second cut feature **234** is positioned closer to the line feature **214** than the opposite sidewall **234B**. Accordingly, the spacing D3 between the sidewall **234A** of the second cut feature **234** and the line feature **214** is smaller than the spacing D4 between the opposite sidewall **234B** of the second cut feature **234** and the line feature **214**. The third cut feature **236** is asymmetrically arranged with respect to a central axis C2 of the line feature **216** extending along the X direction such that the sidewall **236B** of the third cut feature **236** is positioned closer to the line feature **216** than the opposite sidewall **236A**. Accordingly, the spacing D5 between the sidewall **236B** of the third cut feature **236** and the line feature **216** is smaller than the spacing D6 between the opposite sidewall **236A** of the third cut feature **236** and the line feature **216**.

[0034] Although in embodiments illustrated in FIGS. 5A and 5B, both cut features in a pair of adjacent cut features **232** and **236** or **234** and **236** are moved along the positive or negative Y direction to relax the pitch P1 or P2, it is contemplated that the pitch P1 or P2 may be relaxed by moving only one cut feature in the adjacent cut feature pair, e.g., cut features **232** and **236** or cut features **234** and **236** to provide the desired distance between the adjacent cut features, cut features **232** and **236** or cut features **234** and **236**.

[0035] In some embodiments, the narrow spacing between one sidewall **232A**, **234A**, **236A** of each of cut features **232**, **234**, **236** and a corresponding line feature **214**, **216**, however, does not provide adequate cut budget for the subsequently etching process. The cut features **232**, **234**, **236** have to be enlarged to ensure there is enough cut budget for subsequent etching process.

[0036] Referring to FIGS. 1 and 6A-6B, the method **100** proceeds to operation **110**, in which a first angled ion implantation process is performed to implant ions **240** to sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**. FIG. 6A is a top view of the semiconductor structure of FIGS. 5A and 5B after performing the first

angled ion implantation process to implant the ions **240** to the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**, in accordance with some embodiments. FIG. **6B** is a cross-sectional view of the semiconductor structure of FIG. **6A** taken along line B-B'.

[0037] Referring to FIGS. **6A** and **6B**, the ions **240** having a first implantation direction is directed toward the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**. The incident angle θ_1 of the ions **240**, which is defined as an angle between the ions **240** and a normal line perpendicular to the top surface of the substrate **202**, is chosen such that the sidewalls **232B**, **234B**, **236B** on the first side of the cut features **232**, **234**, **236** are exposed to the ions **240**, whereas the sidewalls **232A**, **234A**, **236A** on the second side of the cut features **232**, **234**, **236** are substantially not exposed to the ions **240**. In such a manner, the ion implantation process does not require any additional masking steps, and thus does not incur any additional manufacturing cost. In some embodiments, the incident angle θ_1 can be achieved by placing the substrate **202** on a stage and tilting the stage so as to tilt the substrate **202**. In some embodiments, the stage can be tilted at an angle between about 5 and about 60 degrees. During the first angled ion implantation, the ions **240** strike the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236** to sputter the photoresist away from the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**, causing the localized removal of photoresist from the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**. After the first angled ion implantation, the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236** may be substantially vertical or sloped or have an hourglass profile.

[0038] As a result of the localized photoresist removal from the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**, the cut features **232**, **234**, **236** are enlarged in the positive Y direction such that sidewalls **232B**, **234B**, **236B** of the cut features **232**, **234**, **236** are positioned away from the corresponding line features **214**, **216**. The narrow spacing between the sidewall **236B** of the cut feature **236** and the line feature **216** is thus increased, which allows for easy removal of the unwanted portion of the line feature **216**.

[0039] Exemplary ions **240** that can be employed in the present disclosure include inert gas ions, such as helium, neon, argon, krypton, xenon, and radon ions. In some embodiments, argon ions are used. The ion energy and dose of the ions **240** are adjusted to control the extent of end-push of the cut features **232**, **234**, **236**. The ion energy may be about several hundred eV or greater. In some embodiments, the ions **240** are implanted at an energy from about 0.5 keV to about 20 keV and at a dosage from about 1.0×10^{14} ions/cm² to 8.0×10^{18} ions/cm², although lesser and greater ion energies and doses can be employed.

[0040] The incident angle, energy and/or dosage of the ions **240** may be controlled to control the amount of the photoresist being removed from the sidewalls **232B**, **234B**, **236B** of respective cut features **232**, **234**, **236**.

[0041] Referring to FIGS. **1** and **7A-7B**, the method **100** proceeds to operation **112**, in which a second angled ion implantation process is performed to implant ions **242** to sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**. FIG. **7A** is a top view of the semiconductor structure of FIGS. **6A** and **6B** after performing the second angled ion implantation process to implant the ions **242** to

the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**, in accordance with some embodiments. FIG. **7B** is a cross-sectional view of the semiconductor structure of FIG. **7A** taken along line B-B'.

[0042] Referring to FIGS. **7A** and **7B**, the ions **242** having a second implantation direction is directed toward the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**. The incident angle θ_2 of the ions **242**, which is defined as an angle between the ions **242** and a normal line perpendicular to the top surface of the substrate **202**, is chosen such that the sidewalls **232A**, **234A**, **236A** on the second side of the cut features **232**, **234**, **236** are exposed to the ions **242**, whereas the sidewalls **232B**, **234B**, **236B** on the first side of the cut features **232**, **234**, **236** are substantially not exposed to the ions **242**. In such a manner, the second ion implantation process advantageously does not require any additional masking steps, and thus does not incur any additional manufacturing cost. In some embodiments, the incident angle θ_2 can be achieved by placing the substrate **202** on a stage and tilting the stage so as to tilt the substrate **202**. In some embodiments, the stage can be tilted at an angle between about 5 and about 60 degrees. During the second angled ion implantation, the ions **242** strike the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236** to sputter the photoresist away from the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**, thereby causing the localized removal of photoresist from the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**. After the second angled ion implantation, the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236** may be substantially vertical or sloped or have an hourglass profile.

[0043] As a result of the photoresist removal from the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**, the cut features **232**, **234**, **236** are enlarged in the negative Y direction such that sidewalls **232A**, **234A**, **236A** of the cut features **232**, **234**, **236** are positioned away from the corresponding line features **214**, **216**. Each narrow spacing between the sidewalls **232A**, **234A** of respective cut features **232**, **234** and the line feature **214** is thus increased, which allows for easy removal of the unwanted portions of the line feature **214**.

[0044] Exemplary ions **242** that can be employed in the present disclosure include inert gas ions, such as helium, neon, argon, krypton, xenon, and radon ions. In some embodiments, argon ions are used. The ion energy may be about several hundred eV or greater. In some embodiments, the ions **242** are implanted at an energy from about 0.5 keV to about 20 keV and at a dosage from about 1.0×10^{14} ions/cm² to 8.0×10^{18} ions/cm², although lesser and greater ion energies and doses can be employed.

[0045] The incident angle, energy and/or dosage of the ions **242** may be controlled to control the amount of the photoresist being removed from the sidewalls **232A**, **234A**, **236A** of respective cut features **232**, **234**, **236**. In some embodiments, the condition of the second ion implantation (e.g., incident angle, energy and/or dosage of ions **242**) may be selected to allow more photoresist to be removed by the second angled ion implantation than the first angled ion implantation. In some embodiments, the condition of the second ion implantation (e.g., incident angle, energy and/or dosage of ions **242**) may be selected to allow less photoresist to be removed by the second angled ion implantation than the first angled ion implantation.

[0046] By directional end pushing using the angled implantation processes to enlarge the size of the cut features 232, 234, 236, the cut margin reduced by the pitch relaxation operation is recovered. In some embodiments, the total amount of the end-to-end pushing (i.e., size increase) after the first and second angled implantations can be from about 3 nm to about 8 nm. As a result, the cut process window for the subsequent etching process is improved.

[0047] In some embodiments, the as formed cut features 232, 234, 236 may have an oval shape, which may cause the non-uniform cut in the subsequent cut process. It has been shown that the angled ion implantation processes also help to improve the pattern profile of the cut features 232, 234, 236, and after the ion implantation processes, the profile of the cut features 232, 234, 236 is closer to a rectangular shape which helps to improve the uniformity of the subsequent etching.

[0048] The effects of the angled ion implantation processes on the pattern profiles of exemplary cut features are illustrated in FIG. 9. Curve 1 represents a pattern profile of an oval-shaped cut feature having a critical dimension (CD) of 55 nm without ion implantation treatment. Curve 2 represents a pattern profile of an as formed oval-shaped cut feature having a CD of 55 nm after ion implantation treatment. Curve 3 represents a pattern profile for an oval-shaped cut feature having a CD of 64 nm without ion implantation. Curve 4 shows the simulation results of a rectangular shaped cut pattern having a CD of 64 nm. It can be seen from FIG. 9 that the ion implantation process not only enlarges the CD of the cut feature, it also results in a cut feature with a pattern profile closer to a rectangular shape, which is desirable for achieving uniform etching.

[0049] Referring to FIGS. 1 and 8A-8B, the method 100 proceeds to operation 114, in which portions of the line features 214, 216 that are exposed by the cut features 232, 234, 236 are removed. FIG. 8A is a top view of the semiconductor structure of FIGS. 7A and 7B after removing the portions of the line features 214, 216 that are exposed by the cut features 232, 234, 236, in accordance with some embodiments. FIG. 8B is a cross-sectional view of the semiconductor structure of FIG. 8A taken along line B-B'.

[0050] Referring to FIGS. 8A and 8B, the portions of the line features 214, 216 that are exposed by the cut features 232, 234, 236 are removed using one or more etching processes. In some embodiments, the one or more etching processes include a dry etching process such as a plasma etching process or RIE, a wet etching process, or a combination thereof, so that the corresponding portions of the line features 214, 216 can be selectively etched without impacting the substrate 202. In some embodiments, the etching process includes using etching gases including at least one of carbon tetrafluoride (CF_4), difluoromethane (CH_2F_2), trifluoromethane (CHF_3), other suitable etching gases, or combinations thereof. In some embodiments, the etching process can use solutions such as tetramethylammonium hydroxide (TMAH), hydrofluoric acid (HF), a $\text{HF}/\text{HNO}_3/\text{CH}_3\text{COOH}$ solution, NH_4OH , potassium hydroxide (KOH), or other suitable solutions.

[0051] The line feature 214 is, thus, cut by the cut features 232, 234, forming segmented line features 214A, 214B, 214C spaced apart from each other. Likewise, the line feature 216 is cut by the cut feature 236, forming segmented line features 216A, 216B spaced apart from each other. The

resulting the segmented line features 214A-214C, 216A, 216B and the line features 212, 218 together constitute a target pattern 250.

[0052] After etching, the second photoresist layer 222 is subsequently removed. In some embodiments, the patterned second photoresist layer 222 is removed by a wet stripping process, a plasma ashing process, other suitable methods, and/or combinations thereof. In some embodiments, the plasma ashing process includes using gases including at least one of oxygen (O_2), carbon dioxide (CO_2), nitrogen (N_2), hydrogen (H_2), or combinations thereof.

[0053] In some embodiments, the segmented line features 214A-214C, 216A, 216B, and the line features 212, 218 represent a plurality of fins for formation of FinFETs. In some embodiments, the segmented line features 214A-214C, 216A, 216B, and the line features 212, 218 are dummy features and are used as an etch mask to pattern the substrate 202, thereby forming a plurality of fins in an upper portion of the substrate 202 for formation of FinFETs.

[0054] Related CMOS processing can be employed in various embodiments to continue processing operations and form suitable p-type and n-type FinFET devices that utilize the fins. For example, a first gate structure can be formed over portions of a first set of fins and first source/drain regions comprising dopants of an n-type conductivity can be formed on opposite sides of the first gate structure to provide n-type FinFETs. A second gate structure can be formed over portions of a second set of fins and second source/drain regions comprising dopants of a p-type conductivity can be formed on opposite sides of the second gate structure to provide p-type FinFETs. The gate structures can be formed utilizing a gate first or a gate last process known in the art. Each gate structure may include a gate dielectric and a gate conductor.

[0055] FIGS. 10A-11C are various views of intermediate stages in the formation of the semiconductor structure 200, in accordance with first alternative embodiments. Unless specified otherwise, the materials and the formation processes of the components in the first alternative embodiments are essentially the same as their like components, which are denoted by like reference numerals in the embodiments shown in FIGS. 2A-8B. The details regarding the formation processes and the materials of the components shown in FIGS. 10A-11C are thus found in the discussion of the embodiments shown in FIGS. 2A-8B.

[0056] The initial steps of the first alternative embodiments are substantially similar to those in FIGS. 2A-4B. Operation 102 is performed in which a patterning-target layer 204 is formed over a substrate 202 (FIGS. 2A and 2B). Next, operation 104 is performed in which the patterning-target layer 204 is patterned to form a main pattern 210 including line features 212, 214, 216, 218 (FIGS. 3A and 3B). Next, operation 106 is performed in which a second photoresist layer 220 is deposited over the main pattern 210 (FIGS. 4A and 4B). Next, operation 108 is performed in which the second photoresist layer 220 is patterned to form a patterned second photoresist layer 222 comprising a cut pattern 230.

[0057] FIGS. 10A-10C illustrate various views of the semiconductor structure after operation 108. FIG. 10A is a top view of the semiconductor structure of FIGS. 4A and 4B after patterning the second photoresist layer 220 to form the patterned second photoresist layer 222 comprising the cut pattern 230, in accordance with some embodiments. FIG.

10B is a cross-sectional view of the semiconductor structure of FIG. 10A taken along line B-B'. FIG. 10C is a cross-sectional view of the semiconductor structure of FIG. 10A taken along line C-C'.

[0058] Referring to FIGS. 10A-10C, the cut pattern 230 includes a plurality of cut features, e.g., cut features 232, 234, 236, overlying the main pattern 210. Unlike the embodiments shown in FIGS. 5A and 5B, where all three cut features 232, 234, 236 need to be moved away toward the respective outer line features 212, 218 so as to relax the pitch P1 between the cut features 232 and 236 and the pitch P2 between the cut features 234 and 236, in the first alternative embodiments as shown in FIGS. 10A-10C, only the first and second cut features 232, 234 overlying the line feature 214 need to be moved away in the positive Y direction toward the outer line feature 212 in order to provide a sufficient pitch between each cut feature pair, i.e., cut features 232 and 236, or cut features 234 and 236, which allows for using a single photomask in formation of these cut features 232, 234, 236. Accordingly, the distance D1 between the sidewall 232A of the first cut feature 232 and the line feature 214, and the distance D3 between the sidewall 234A of the second cut feature 234 and the line feature 214 remain small, which does not provide enough cut budget for the subsequent etching process. In contrast, the distance D5 between the sidewall 236B of the third cut feature 236 and the line feature 216, which is greater than the distance D1 or D3, is sufficiently large to provide enough cut budget for the subsequent etching process. Accordingly, only a single ion implantation process is required to push the end of each of cut feature 232, 234 in the negative Y direction so that the sidewalls 232A, 234A of respective cut features 232, 234 can be further away from the line feature 214. No ion implantation process is needed to push the end of the cut feature in the positive Y direction because the sidewall 236B of the cut feature 236 is sufficiently away from the line feature 216. Operation 110 is thus omitted.

[0059] Next, operation 112 is performed in which an angled ion implantation process is performed to implant ions 242 to sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236. FIG. 11A is a top view of the semiconductor structure of FIGS. 10A-10C after performing the angled ion implantation process to implant the ions 242 to the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, in accordance with some embodiments. FIG. 11B is a cross-sectional view of the semiconductor structure of FIG. 11A taken along line B-B'. FIG. 11C is a cross-sectional view of the semiconductor structure of FIG. 11A taken along line C-C'.

[0060] Referring to FIGS. 11A-11C, the ions 242 are directed toward the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236. The incident angle of the ions 242, which is defined as an angle between the ions 242 and a normal line perpendicular to the top surface of the substrate 202, is chosen such that the sidewalls 232A, 234A, 236A on the second side of the cut features 232, 234, 236 are exposed to the ions 242, whereas the sidewalls 232B, 234B, 236B on the first side of the cut features 232, 234, 236 are substantially not exposed to the ions 242. In such a manner, the ion implantation process advantageously does not require any additional masking steps, and thus does not incur any additional manufacturing cost. During the angled ion implantation, the ions 242 strike the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236 to sputter the

photoresist away from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, thereby causing the localized removal of photoresist from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236.

[0061] As a result of the photoresist removal from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, the cut features 232, 234, 236 are enlarged in the negative Y direction such that sidewalls 232A, 234A, 236A of the cut features 232, 234, 236 are positioned away from the corresponding line features 214, 216. Each narrow spacing between the sidewalls 232A, 234A of respective cut features 232, 234 and the line feature 214 is thus increased, which allows for easy removal of the unwanted portions of the line feature 214.

[0062] By directional end pushing using the angled implantation process to enlarge the size of the cut features 232, 234, the cut margin reduced by the pitch relaxation operation is recovered. As a result, the cut process window for the subsequent etching process is improved.

[0063] Next, operation 114 is performed, in which portions of the line features 214, 216 that are exposed by the cut features 232, 234, 236 are removed to provide the semiconductor structure 200 (FIGS. 8A and 8B).

[0064] Related CMOS processing can be employed in various embodiments to continue processing operations and form suitable p-type and n-type FinFET devices that utilize the fins. For example, a first gate structure can be formed over portions of a first set of fins and first source/drain regions comprising dopants of an n-type conductivity can be formed on opposite sides of the first gate structure to provide n-type FinFETs. A second gate structure can be formed over portions of a second set of fins and second source/drain regions comprising dopants of a p-type conductivity can be formed on opposite sides of the second gate structure to provide p-type FinFETs. The gate structures can be formed utilizing a gate first or a gate last process known in the art. Each gate structure may include a gate dielectric and a gate conductor.

[0065] FIGS. 12A-14C are various views of intermediate stages in the formation of the semiconductor structure 200, in accordance with second alternative embodiments. Unless specified otherwise, the materials and the formation processes of the components in the second alternative embodiments are essentially the same as their like components, which are denoted by like reference numerals in the embodiments shown in FIGS. 2A-8B. The details regarding the formation processes and the materials of the components shown in FIGS. 12A-14C are thus found in the discussion of the embodiments shown in FIGS. 2A-8B.

[0066] The initial steps of the second alternative embodiments are substantially similar to those in FIGS. 2A-4B. Operation 102 is performed in which a patterning-target layer 204 is formed over a substrate 202 (FIGS. 2A and 2B). Next, operation 104 is performed in which the patterning-target layer 204 is patterned to form a main pattern 210 including line features 212, 214, 216, 218 (FIGS. 3A and 3B). Next, operation 106 is performed in which a second photoresist layer 220 is deposited over the main pattern 210 (FIGS. 4A and 4B). Next, operation 108 is performed in which the second photoresist layer 220 is patterned to form a patterned second photoresist layer 222 comprising a cut pattern 230.

[0067] FIGS. 12A-12C illustrate various views of the semiconductor structure after operation 108. FIG. 12A is a

top view of the semiconductor structure of FIGS. 4A and 4B after patterning the second photoresist layer 220 to form the patterned second photoresist layer 222 comprising the cut pattern 230, in accordance with some embodiments. FIG. 12B is a cross-sectional view of the semiconductor structure of FIG. 12A taken along line B-B'. FIG. 12C is a cross-sectional view of the semiconductor structure of FIG. 12A taken along line C-C'.

[0068] Referring to FIGS. 12A-12C, the cut pattern 230 includes a plurality of cut features, e.g., cut features 232, 234, 236, overlying the main pattern 210. The cut features 232, 234, 236 are positioned to relax the pitch requirements such that they can be fabricated using a single photomask. For example, as shown in FIG. 12A, the cut features 232, 234, 236 are positioned such that the distance S1 between the sidewall 232B of the first cut feature 232 and the adjacent line feature 212 is less than the distance S2 between the sidewall 236A of the third cut feature 236 and the adjacent line feature 218, while the distance S3 between the sidewall 232A of the first cut feature 232 and the line feature 216 is greater than the distance S4 between the sidewall 236B and the line feature 214. Accordingly, such cut feature arrangement causes the space in the positive Y direction less than the space in the negative Y direction, and thus asymmetric implantation can be used to enlarge the sizes of respective cut features 232, 234, 236.

[0069] Next, operation 110 is performed, in which a first angled ion implantation process is performed to implant ions 240 to sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236 (FIGS. 13A-13C). FIG. 13A is a top view of the semiconductor structure of FIGS. 12A-12C after performing the first angled ion implantation process to implant the ions 240 to the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236, in accordance with some embodiments. FIG. 13B is a cross-sectional view of the semiconductor structure of FIG. 13A taken along line B-B'. FIG. 13C is a cross-sectional view of the semiconductor structure of FIG. 13A taken along line C-C'.

[0070] Referring to FIGS. 13A-13C, the ions 240 having a first implantation direction are directed toward the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236. The incident angle of the ions 240, which is defined as an angle between the ions 240 and a normal line perpendicular to the top surface of the substrate 202, is chosen such that the sidewalls 232B, 234B, 236B on the first side of the cut features 232, 234, 236 are exposed to the ions 240, whereas the sidewalls 232A, 234A, 236A on the second side of the cut features 232, 234, 236 are substantially not exposed to the ions 240. During the first angled ion implantation, the ions 240 strike the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236, to sputter the photoresist away from the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236, thereby causing the localized removal of photoresist from the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236.

[0071] Because the spacing between the sidewall 232A of the first cut feature 232 and the underlying line feature 216 is greater than the spacing between the sidewall 236B of the third cut feature 236 and the line feature 214, i.e., $S3 > S4$, less photoresist needs to be removed from the sidewall 236B of the third cut feature 236 to provide enough cut budget for removal of the unwanted portion of the line feature 216. Accordingly, the energy and/or dosage of the ions 240 are controlled such that a relatively small portion of the photo-

resist is removed from each of the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236. In some embodiments, each removed photoresist portion has a triangular cross-sectional shape with a base length $\Delta L1$. After the first angled ion implantation, each sidewall 232B, 234B, 236B of the cut features 232, 234, 236 is inclined at a first angle $\beta1$ with respect to the normal line that is perpendicular to the top surface of the substrate 202.

[0072] As a result of the photoresist removal from the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236, the cut features 232, 234, 236 are enlarged in the positive Y direction such that sidewalls 232B, 234B, 236B of the cut features 232, 234, 236 are positioned away from the corresponding line features 214, 216. The spacing between the sidewall 236B of the cut feature 236 and the line feature 216 is thus increased, which allows for easy removal of the unwanted portion of the line feature 216.

[0073] Next, operation 112 is performed, in which a second angled ion implantation process is performed to implant ions 242 to sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236 (FIGS. 14A-14C). FIG. 14A is a top view of the semiconductor structure of FIGS. 13A-13C after performing the second angled ion implantation process to implant the ions 242 to the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, in accordance with some embodiments. FIG. 14B is a cross-sectional view of the semiconductor structure of FIG. 14A taken along line B-B'. FIG. 14C is a cross-sectional view of the semiconductor structure of FIG. 14A taken along line C-C'.

[0074] Referring to FIGS. 14A-14C, the ions 242 having a second implantation direction is directed toward the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236. The incident angle of the ions 242, which is defined as an angle between the ions 242 and a normal line perpendicular to the top surface of the substrate 202, is chosen such that the sidewalls 232A, 234A, 236A on the second side of the cut features 232, 234, 236 are exposed to the ions 242, whereas the sidewalls 232B, 234B, 236B on the first side of the cut features 232, 234, 236 are substantially not exposed to the ions 242. During the second angled ion implantation, the ions 242 strike the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236 to sputter the photoresist away from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, thereby causing the localized removal of photoresist from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236.

[0075] In the second alternative embodiments, because the distance S3 between the sidewall 232A of the first cut feature 232 and the adjacent line feature 216 is greater than the distance S4 between the sidewall 236B of the third cut feature 236 and the adjacent line feature 214, i.e., $S3 > S4$, more photoresist can be removed from the sidewall 232A of the first cut feature 232 to provide enough cut budget for removal of the unwanted portion of the line feature 214 underlying the first cut feature 232. Accordingly, the energy and/or dosage of the ions 242 are controlled to be higher than the energy and/or dose of the ions 240. A relatively large amount of the photoresist thus is removed from each of the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236 compared with the amount of the photoresist that is removed from each of the sidewalls 232B, 234B, 236B of respective cut features 232, 234, 236. In some embodiments, each removed photoresist portion has a tri-

angular cross-sectional shape with a base length ΔL_2 , which is greater than ΔL_1 . After the second angled ion implantation, each sidewall 232A, 234A, 236A of the cut features 232, 234, 236 are inclined at a second angle β_2 with respect to the normal line that is perpendicular to the top surface of the substrate 202. The second incline angle β_2 is greater than the first incline angle β_1 .

[0076] As a result of the photoresist removal from the sidewalls 232A, 234A, 236A of respective cut features 232, 234, 236, the cut features 232, 234, 236 are enlarged in the negative Y direction such that sidewalls 232A, 234A, 236A of the cut features 232, 234, 236 are positioned away from the corresponding line features 214, 216. Each narrow spacing between the sidewalls 232A, 234A of respective cut features 232, 234 and the line feature 214 is thus increased, which allows for easy removal of the unwanted portions of the line feature 214.

[0077] Next, operation 114 is performed, in which portions of the line features 214, 216 that are exposed by the cut features 232, 234, 236 are removed to provide the semiconductor structure 200 (FIGS. 8A and 8B).

[0078] Related CMOS processing can be employed in various embodiments to continue processing operations and form suitable p-type and n-type FinFET devices that utilize the fins. For example, a first gate structure can be formed over portions of a first set of fins and first source/drain regions comprising dopants of an n-type conductivity can be formed on opposite sides of the first gate structure to provide n-type FinFETs. A second gate structure can be formed over portions of a second set of fins and second source/drain regions comprising dopants of a p-type conductivity can be formed on opposite sides of the second gate structure to provide p-type FinFETs. The gate structures can be formed utilizing a gate first or a gate last process known in the art. Each gate structure may include a gate dielectric and a gate conductor.

[0079] One aspect of this description relates to a method for forming a semiconductor structure. The method includes forming a pattern comprising a first line feature and a second line feature extending in a first direction on a substrate, depositing a photoresist layer on the substrate to cover the pattern, patterning the photoresist layer to form a cut pattern comprising a first cut feature exposing a portion of the first line feature and a second cut feature exposing a portion of the second line feature. In a top view, at least one of the first cut feature and the second cut feature is asymmetrically arranged with respect to a central axis of a corresponding first or second line feature. The method further includes performing at least one angled ion implantation to enlarge the first cut feature and the second cut feature in at least one direction perpendicular to the first direction, removing the portion of the first line feature exposed by the first cut feature and the portion of the second line feature exposed by the second cut feature, and removing the photoresist layer.

[0080] Another aspect of this description relates to a method for forming a semiconductor structure. The method includes forming a pattern on a substrate. The pattern includes a first line feature and a second line feature extending in a first direction; and the method further includes depositing a photoresist layer on the substrate to cover the pattern. The method further includes patterning the photoresist layer to form a first cut feature exposing a portion of the first line feature and a second cut feature exposing a portion of the second line feature. In a top view, the first cut

feature is asymmetrically arranged with respect to a central axis of the first line feature. The method further includes performing a first angled ion implantation to enlarge the first cut feature and the second cut feature in a second direction perpendicular to the first direction, removing the portion of the first line feature exposed by the first cut feature and the portion of the second line feature exposed by the second cut feature, and removing the photoresist layer.

[0081] Still another aspect of this description relates to a method for forming a semiconductor structure. The method includes etching a patterning-target layer to form a pattern on a substrate. The pattern includes a first line feature and a second line feature extending in a first direction; and the method further includes depositing a photoresist layer on the substrate to cover the pattern. The method further includes patterning the photoresist layer to form a first cut feature exposing a portion of the first line feature and a second cut feature exposing a portion of the second line feature. Each of the first cut feature and the second cut features has first and second sidewalls extending in the first direction. A first spacing between the first sidewall of the first cut feature and the first line feature is greater than a second spacing between the second sidewall of the first cut feature and the first line feature. A first spacing between the first sidewall of the second cut feature and the second line feature is less than a second spacing between the second sidewall of the second cut feature and the second line feature. The second sidewall of the first cut feature is proximate to the second line feature, and the first sidewall of the second cut feature is proximate to the first line feature; and the method further includes performing a first angled ion implantation to enlarge the first cut feature and the second cut feature in a second direction perpendicular to the first direction. The method further includes performing a second angled ion implantation to enlarge the first cut feature and the second cut feature in a third direction opposite the second direction, removing the portion of the first line feature exposed by the first cut feature and the portion of the second line feature exposed by the second cut feature, and removing the photoresist layer.

[0082] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method for forming a semiconductor device, comprising:

forming a plurality of line features extending in a first direction on a substrate, the plurality of line features comprising a first line feature, a second line feature adjacent to and spaced from the first line feature by a first spacing, a third line feature adjacent to and spaced from the second line feature by a second spacing, and a fourth line feature adjacent to and spaced from the

- third line feature by a third spacing, wherein the second spacing is less than each of the first spacing and the third spacing;
- depositing a photoresist layer over the substrate and the plurality of line features;
- forming a patterned photoresist layer comprising a plurality of trenches, the plurality of trenches comprising a first trench on the second line feature and a second trench on the third line feature, wherein in a top view, the first trench and the second trench are asymmetrically arranged with respect to central axes of the second line feature and the third line feature, respectively;
- implanting first ions onto a first sidewall of the first trench that is located between the second and third line features and a first sidewall of the second trench that is located between the third and fourth line features, thereby enlarging a spacing between the first sidewall of the first trench and the second line feature and a spacing between the first sidewall of the second trench and the third line feature in a second direction perpendicular to the first direction;
- removing a portion of the second line feature exposed by the first trench and a portion of the third line feature exposed by the second trench; and
- removing the patterned photoresist layer.
2. The method of claim 1, further comprising forming transistors on the remaining portions of the second and third line features.
3. The method of claim 1, wherein the first ions comprise argon ions.
4. The method of claim 1, wherein the first ions are implanted at an energy from about 0.5 keV to about 20 keV and/or at a dosage from about 1.0×10^{14} ions/cm² to 8.0×10^{18} ions/cm².
5. The method of claim 1, further comprising implanting second ions onto a second sidewall of the first trench that is opposite the first sidewall and located between the first and second line features, and onto a second sidewall of the second trench that is opposite the first sidewall and located between the second and third line features, thereby enlarging a spacing between the second sidewall of the first trench and the second line feature and a spacing between the second sidewall of the second trench and the third line feature in a third direction opposite to the second direction.
6. The method of claim 5, wherein the first ions are implanted toward the first sidewall of each of the first and second trenches at a first incident angle, and the second ions are implanted toward the second sidewall of each of the first and second trenches at a second incident angle different from the first incident angle.
7. The method of claim 5, wherein the second ions are implanted at an energy higher than the energy of the first ions.
8. The method of claim 5, wherein the second ions are implanted at a dose higher than the dose of the first ions.
9. The method of claim 5, wherein after implanting the first ions, the first sidewall is inclined at a first angle, and after implanting the second ions, the second sidewall is inclined at a second angle greater than the first angle.
10. The method of claim 1, wherein each of the first and fourth line features has a width of 15.5 nm, and each of the second and third line features has a width of 9 nm.
11. The method of claim 1, wherein each of the first and third spacings is 39 nm, and the second spacing is 32 nm.

12. The method of claim 1, further comprising:
- etching an upper portion of the substrate to form a plurality of fins using the first and fourth line features and the remaining portions of the second and third line features as an etch mask; and
- forming transistors on the plurality of fins.
13. A method for forming a semiconductor device, comprising:
- forming a plurality of line features extending in a first direction on a substrate, the plurality of line features comprising a first line feature, a second line feature adjacent to and spaced from the first line feature by a first spacing, a third line feature adjacent to and spaced from the second line feature by a second spacing, and a fourth line feature adjacent to and spaced from the third line feature by a third spacing, wherein the second spacing is less than each of the first spacing and the third spacing;
- depositing a photoresist layer over the substrate and the plurality of line features;
- forming a patterned photoresist layer comprising a plurality of trenches, the plurality of trenches comprising a first trench on the second line feature and a second trench on the third line feature, wherein in a top view, the first and second trenches are asymmetrically arranged with respect to central axes of the second and third line features, respectively;
- performing a first ion implantation to implant first ions onto a first sidewall of the first trench that is located between the second and third line features and a first sidewall of the second trench that is located between the third and fourth line features, thereby enlarging a spacing between the first sidewall of the first trench and the second line feature and a spacing between the first sidewall of the second trench and the third line feature in a second direction perpendicular to the first direction;
- performing a second ion implantation to implant second ions onto a second sidewall of the first trench that is opposite the first sidewall and located between the first and second line features, and onto a second sidewall of the second trench that is opposite the first sidewall and located between the second and third line features, thereby enlarging a spacing between the second sidewall of the first trench and the second line feature and a spacing between the second sidewall of the second trench and the third line feature in a third direction opposite to the second direction;
- removing a portion of the second line feature exposed by the first trench and a portion of the third line feature exposed by the second trench;
- removing the patterned photoresist layer; and
- forming transistors on the remaining portions of the second and third line features.
14. The method of claim 13, wherein the first ion implantation is performed at a first incident angle, and the second ion implantation is performed at a second incident angle which is the same as or different than the first incident angle.
15. The method of claim 14, wherein the first incident angle and the second incident angle are achieved by placing the substrate on a stage and tilting the stage.
16. The method of claim 15, wherein the stage is tilted at an angle from 5 to 60 degrees.

17. The method of claim **16**, wherein a pitch between the first trench and the second trench is greater than a minimum pitch that can be obtained with a single lithography process at a given technology node.

18. A method for forming a memory device, comprising: forming a plurality of line features extending in a first direction on a substrate, the plurality of line features comprising a first line feature, a second line feature adjacent to and spaced from the first line feature by a first spacing, a third line feature adjacent to and spaced from the second line feature by a second spacing, and a fourth line feature adjacent to and spaced from the third line feature by a third spacing, wherein the second spacing is less than each of the first spacing and the third spacing;

depositing a photoresist layer over the substrate and the plurality of line features;

forming a patterned photoresist layer comprising a plurality of trenches, the plurality of trenches comprising a first trench on the second line feature and a second trench on the third line feature, wherein a first sidewall of the first trench that is located between the second and third line features is spaced from the second line feature by a first distance, a second sidewall of the first trench that is located between the first and second line features

is spaced from the second line feature by a second distance greater than the first distance, a first sidewall of the second trench that is located between the third and fourth line features is spaced from the third line feature by a third distance, and a second sidewall of the second trench located between the second and third line features is spaced from the third line feature by a fourth distance;

performing ion implantation to implant ions onto the first sidewall of each of the first trench and the second trench, thereby enlarging the first distance and the third distance;

removing a portion of the second line feature exposed by the first trench and a portion of the third line feature exposed by the second trench; and

removing the patterned photoresist layer.

19. The method of claim **18**, further comprising forming transistors on the remaining portions of the second and third line features.

20. The method of claim **18**, further comprising performing another ion implantation to implant ions onto the second sidewall of each of the first trench and the second trench to enlarge the second distance and the fourth distance.

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